Sanskrit to English Neural Machine Translation

Natural Language Understanding - Assignment 2
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github.com/G25AIT1012/Natural-Language-Understanding-Assignment-2

1. Introduction

1. In this assignment I'm building a system that translates Sanskrit into English.
2. Difficult to translate because of complicated grammar and lack of translated literature for Sanskrit.
3. Training a model from scratch with just 10,000 pairs was not very effective so I fine tuned a pre-trained multilingual model instead.
4. I have used Meta's NLLB-200 (distilled, 600M parameters), which is already supporting both Sanskrit and English.
5. This dataset is used for fine tuning.
6. On the test set it was able to achieve a corpus BLEU of 0.2615 and a BERTScore (F1) of 0.6343
7. All pretrained models are listed in Section 8, and no external APIs were used.

2. Architecture

NLLB-200 is an encoder–decoder Transformer. I did not change the architecture; I only fine-tuned it.

1. **Encoder**: reads the Sanskrit sentence using multi-head self-attention and feed-forward layers, and turns it into contextual vectors.
2. **Decoder:** It creates the English translation one token at a time, by using masked self-attention over the tokens it has already created, cross-attention over the encoder output, and a feed-forward layer.
3. **Attention:** This allows each word to attend to the others, which is useful for Sanskrit's word agreement and long-distance dependencies.
4. **Language and decoding:** I set the source language to san_Deva and force the first output token to eng_Latn so the model generates English. Decoding uses beam search (beam 6, length penalty 1.0, no-repeat n-gram 3).

3. Training Procedure

1. **Preprocessing:** this is to align the Sanskrit and English files on Source_id and removed empty rows.
2. **Tokenization:** For this I used NLLB's own SentencePiece tokenizer (max length 128 tokens).
3. **Hyperparameters:**
   - epochs=10
   - learning rate=2e^-5
   - batch size=8 (with gradient accumulation 2 -> effective 16)
   - warmup ratio=0.10
4. **Optimization:**
   - Trained using Hugging Face's -> Seq2SeqTrainer with AdamW optimizer and a learning rate warmup followed by decay.
   - Employed the mixed-precision training (fp16) and gradient checkpointing to lower GPU memory consumption.
5. **Regularization:** label smoothing 0.10, weight decay 0.01, and early stopping (patience 2); the best checkpoint on dev BLEU was kept.

4. Results

1. Results are on the test set (1,000 sentences).
2. BLEU uses NLTK with default weights; BERTScore uses the official bert-score library (F1, rescaled with baseline).

Metric	Value (test set)
Corpus BLEU	0. 2615
Sentence BLEU	0.2301
BERTScore (F1)	0.6343
Inference time (1,000 sentences)	94.64 seconds
Model parameters	615,073,792

3. Inference ran on a single NVIDIA A100 (40 GB) GPU, using about 16.8 GB of GPU memory.
4. The dev set was used during training to pick the best checkpoint.
5. Training and validation loss went down each epoch while dev BLEU went up, until early stopping; the full log is in the notebook.
6. BLEU (~0.26) is much lower than BERTScore (~0.63).
7. This is because the model often gives a correct translation using different words than the reference, which BLEU punishes but BERTScore does not.
8. The examples below explain this.

4.1 Training Curves

The plots below show the training and validation loss and the validation BLEU across epochs.

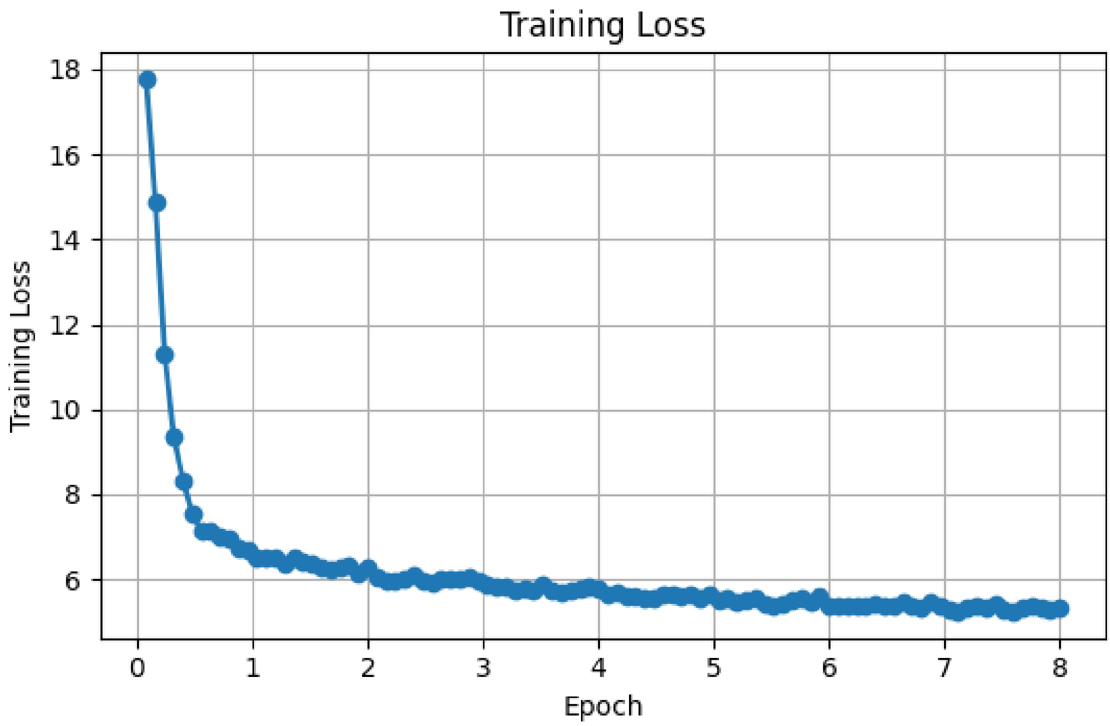

Figure 1. Training Loss over epochs.

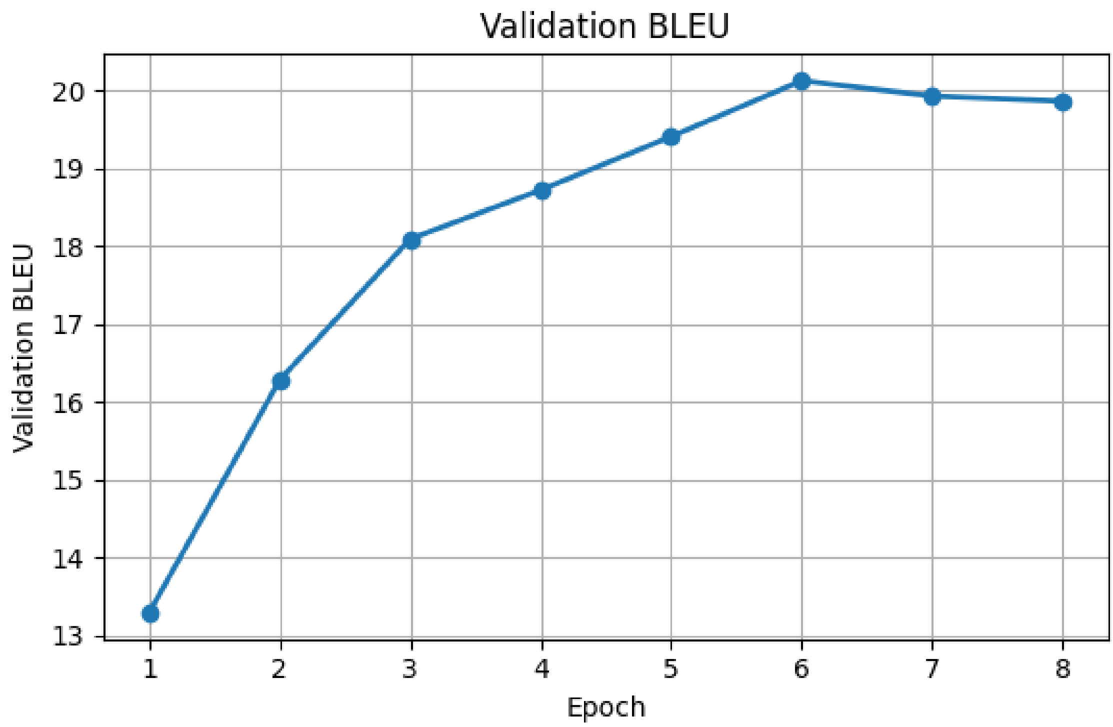

Figure 2. Validation BLEU over epochs.

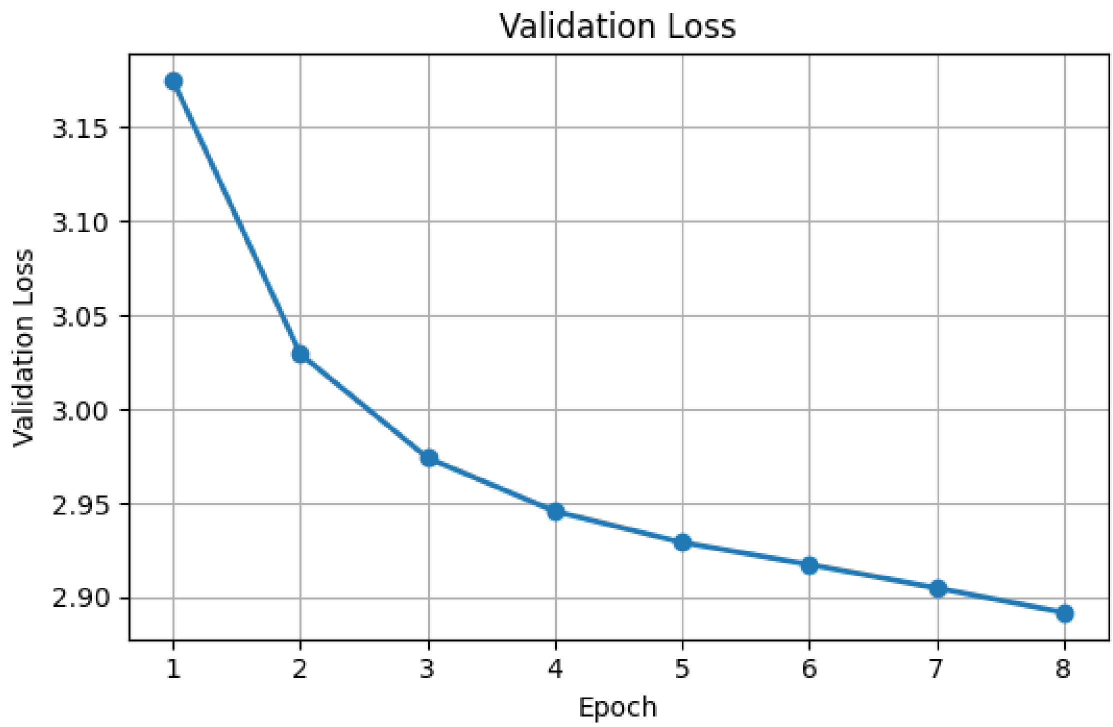

Figure 3. Validation Loss over epochs.

5. Translation Examples and Error Analysis

Representative examples from the test-set predictions:

ID	Source (Sanskrit)	Reference (English)	Prediction (English)
28	बाल: त्वयि प्रेमं प्रदर्शयति ।	Boy displays affection in you.	Boy displays affection in you.
131	यह पूरे शरीर में रक्त परिसंचरण में सुधार करता है ।	It improves blood circulation throughout the body.	It improves blood circulation throughout the body.

3	तदा, तत्स्वयं ड्रैवर निमित्तम् अन्वेष्यति ।	Then it will automatically begin searching for drivers.	Then it will automatically search for the driver.
4	सर्वेभ्यः इटरेशन्-अर्थम्, iterator इतीदं प्रत्येकस्मै इण्डेक्स्-वेल्यू-इत्यस्मै सेट् क्रियते । 1,1 पश्चात् 1,2 एवम्...	The iterator will be set to each of the indices values...	The meaning of iteration for all iterators is set to the index value...
401	भिल्लमाचार्य इति अस्यनामान्तरम्।	He was also known as Bhilacharya.	His name is Villamacharya.
867	डौन्-लोड्-विषये विवरणम्:	Information regarding download:	Details of the download
971	श्वास भरें, बिना किसी दबाव के श्वास भरें । यह श्वास सामान्य रूप से भरा जाना चाहिए ।	Inhale spontaneously and passively. Let air enter the body through the passive inhalation.	Inhale, inhale without any pressure. This inhalation should be filled normally.

1. **Best cases:** short and simple sentences (28, 131) are translated exactly
2. **Common issue:** The model is often able to preserve the correct meaning while using different wording. In the above example-> the ID 3 translates "begin searching for drivers" as "search for the driver", and the ID 4 paraphrases the iterator description but preserves the underlying meaning. These semantic variations reduce BLEU even though fluent translations are produced.
3. **Weak spot:** Proper nouns and transliterated names are still a problem. In ID 401, Bhilacharya is translated as Villamacharya, this shows that transliteration errors are more common than grammatical errors.
4. **BLEU limitation:** BLEU penalizes valid paraphrasing also. For example, ID 867 translates "Information regarding download" as "Details of the download". The wording is different, the meaning is essentially the same. Similarly, ID 971 also correctly conveys the breathing instruction using a more natural English expression. These cases receive lower BLEU despite being semantically accurate, which is why BERTScore is also reported.

6. Experiments

- **Decoding.** I have tuned the decoding parameters to improve the quality of the translation. Beam search with a beam size of 6, a length penalty of 1.0, and a maximum decode length of 256 produced more complete and fluent translations than the default decoding settings.
- **Training.** I fine-tuned the pretrained NLLB-200 model with a learning rate of 2×10^{-5}, batch size 8 with gradient accumulation 2, label smoothing 0.1, and weight decay 0.01. The model converges in a few epochs, and early stopping selects the checkpoint with the highest validation BLEU.
- **Evaluation:** I used - Corpus BLEU, Sentence BLEU and BERTScore (F1) for evaluating the translation quality. BLEU measures n-gram overlap with the reference translations, while BERTScore captures semantic similarity, making it a useful complementary metric for paraphrased translations.

7. Discussion

- **Design choices:** I fine-tuned NLLB-200 Distilled (600M) model instead of training from scratch because it already knows Sanskrit and gives much better results on this small dataset. The increase in computation cost is due to the size: at ~615M parameters it is large and slower, which reduces the efficiency score.
- **Challenges:** the data mixes everyday sentences with software and technical terms, and Sanskrit's rich word forms and single references make BLEU look low even when translations are good.
- **Limitations:** only the given data was used (no extra data), the model is large, and BLEU understates real quality - which is why BERTScore is reported too. A smaller model or a method like LoRA could reduce the cost.

8. Pre-trained Models Disclosure

- **Translation model:** facebook/nllb-200-distilled-600M, pretrained by Meta and fine-tuned here on the provided data only (a model download, not an API).
- **Tokenizer:** NLLB's own SentencePiece tokenizer, used as-is.
- **BERTScore:** the bert-score library uses a pretrained RoBERTa for evaluation only, it was not used during model training or inference.
- **APIs:** No external translation or inference APIs have been used for this project.

9. References

1. NLLB Team, No Language Left Behind: Scaling Human-Centered Machine Translation, Meta AI, 2022.
2. Vaswani, A., et al., Attention Is All You Need, Advances in Neural Information Processing Systems (NeurIPS), 2017.
3. Fan, A., et al., Beyond English-Centric Multilingual Machine Translation (M2M-100), Journal of Machine Learning Research (JMLR), 2021.
4. Wolf, T., et al., Transformers: State-of-the-Art Natural Language Processing, EMNLP System Demonstrations, 2020.
5. Papineni, K., Roukos, S., Ward, T., & Zhu, W., BLEU: A Method for Automatic Evaluation of Machine Translation, ACL, 2002.
6. Zhang, T., et al., BERTScore: Evaluating Text Generation with BERT, ICLR, 2020.

10. Code and Submission

- **Code in GitHub:** https://github.com/G25AIT1012/Natural-Language-Understanding-Assignment-2
- **Repository content:**
    1. Code.ipynb – The training and inferencing notebook.
    2. Report.pdf - Assignment report.
    3. submission.csv - Final predictions on the test set
- **Fine-tuned model:** The trained model (finetuned_model.zip) is also uploaded separately in the following Google Drive link:
  https://drive.google.com/drive/u/5/folders/1BDJzZ6UUdAqvga8ipGztb95kWjRX2zND